

**Time:** 6 minutes**Outcome:** Calculate private, public, and national saving in closed and open economies.**Difficulty:** ●●○ Intermediate

Saving, Investment, and National-Saving Identities



THE CORE IDEA

With net taxes T , private saving is $Y - T - C$ and public saving is $T - G$. National saving is their sum: $S = Y - C - G$. In a closed economy, $S = I$. In an open economy, $S = I + NCO$; because $NX = NCO$ in the course model, $S = I + NX$. These are aggregate identities, not claims that each saver directly funds one investor.



HOW TO RECOGNIZE IT

- Budget surplus means $T - G$ is positive; a deficit makes public saving negative.
- Adding private and public saving cancels T and gives $Y - C - G$.
- Use $S = I$ only for the closed-economy model.
- In an open economy, saving can finance domestic investment or net capital outflow.



WATCH OUT

Do not reverse the public-saving sign. $T - G$ is public saving; $G - T$ is the budget deficit. Also state the closed-economy assumption before writing $S = I$.



WORKED EXAMPLE: FROM COMPONENTS TO OPEN-ECONOMY SAVING

KEY RELATIONSHIPS

- Private saving = $Y - T - C$
- Public saving = $T - G$
- National saving = $Y - C - G$
- Closed: $S = I$
- Open: $S = I + NCO$; $NX = NCO$

Let $Y = \$1,200$, $T = \$150$, $C = \$760$, and $G = \$260$. Private saving is \$290; public saving is -\$110; national saving is \$180. If domestic investment is \$140, then $NCO = S - I = \$40$. The accounting identity also gives $NX = \$40$.



CHECK YOURSELF

Private saving is \$220 and public saving is -\$50. What is national saving?

**READY?**

Return to the game and master this concept.